# Detecting Power Plant $NO_x$ Emissions Changes using Physics-Guided Deep Learning and TEMPO Satellite Imagery

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## Abstract

NASA's Tropospheric Emissions: Monitoring of Pollution (TEMPO) instrument provides hourly daytime observations of tropospheric nitrogen dioxide ($NO_2$) across North America. These observations could support independent, facility-level monitoring of power plant nitrogen oxides ($NO_x$) emissions, filling gaps caused by continuous emissions monitoring system (CEMS) failures and covering facilities not subject to hourly emissions-reporting requirements. Previous studies have estimated hourly point-source emissions from individual satellite scenes using empirical methods such as cross-sectional flux and the directional derivative approach. However, these estimates are sensitive to retrieval uncertainty, changing atmospheric transport, TEMPO's coarse spatial resolution, signals from nearby emitters, and temporal mismatch with hourly CEMS totals. We instead formulate facility monitoring as change detection over sequences of TEMPO observations. This relative target uses repeated scans to capture plume evolution while reducing sensitivity to persistent background levels and biases in absolute column values. We develop a physics-guided convolutional gated recurrent unit model (ConvGRU) that processes five consecutive TEMPO observations, High-Resolution Rapid Refresh (HRRR) meteorological fields, and transport residuals. The network jointly classifies the target power plant's emissions as increasing, steady, or decreasing and predicts a spatial map of contributions from emitting facilities within the observed scene. On a held-out evaluation set containing more than 20,000 samples from over 95 U.S. power plants observed between August 2023 and July 2026, the model achieves a classification macro-F1 score of 0.54, which outperformed a baseline MLP (0.49). These results support the use of sequential TEMPO imagery for detecting short-term changes in power plant emissions across facilities and observing conditions.

## 1 Introduction

Based on U.S. Environmental Protection Agency (EPA) inventory data, power plants in the U.S. electricity-generation sector emitted approximately 0.74 million short tons of $NO_x$ in 2023, about 9.5% of national emissions [13]. $NO_x$ emissions contribute to ground-level ozone and secondary particulate matter, pollutants that can inflame and damage the airways, reduce lung function, and aggravate asthma [11]. EPA regulations require most large U.S. fossil-fuel-fired generating units that produce electricity for sale to submit unit-level hourly emissions data on a quarterly basis [14]. However, monitoring equipment malfunctions and maintenance can make CEMS measurements unavailable or invalid, requiring plants to report calculated substitute values instead of direct measurements [12]. Behind-the-meter generators that do not sell electricity can fall outside federal hourly reporting requirements, as illustrated by the natural-gas turbines powering SpaceXAI's Colossus 2 data center [5]. Outside the United States, comparable public hourly, unit-level emissions data are generally unavailable, even where facilities continuously monitor emissions [6]. Independent estimates could therefore fill reporting gaps and reveal short-term changes before the EPA's quarterly release of hourly emissions data.

TEMPO is an ultraviolet-visible spectrometer hosted aboard a geostationary satellite that observes tropospheric vertical column densities (TVCDs) for various trace gases across North America during daylight hours [16]. Earlier polar-orbiting instruments, including the Ozone Monitoring Instrument (OMI) and TROPOspheric Monitoring Instrument (TROPOMI), typically observe a given location once per day; TEMPO's geostationary view provides approximately hourly scans with finer spatial resolution [7, 15, 16]. Hourly sampling captures intraday changes in power plant plumes and atmospheric transport that daily observations can miss. However, power plant signals in TEMPO $NO_2$ TVCDs can be weakened or obscured by factors including evolving meteorology and atmospheric chemistry, cloud cover, viewing geometry and retrieval uncertainty, dilution within TEMPO's 2.0-by-4.75 km spatial resolution per pixel, and interference from nearby emission sources [5, 6, 10]. Cross-comparisons with CEMS data, which is the most reliable ground truth available, introduce an additional temporal mismatch because TEMPO observes a facility at a specific time during each hourly scan, whereas CEMS data aggregate emissions over an entire clock hour [10, 14].

Satellite-based point-source studies typically combine observed column enhancements with modeled atmospheric transport. Cross-sectional flux (CSF) integrates background-subtracted columns across plume transects, while the directional derivative approach (DDA) relates spatial column gradients to wind fields [8, 9]. Santaren et al. benchmarked five such methods on synthetic daily satellite scenes with known source rates and found that cloud gaps and wind errors limited their accuracy and coverage [8]. For TEMPO, Sun et al. added hourly changes in the observed $NO_2$ column to DDA and compared estimates for 14 U.S. power plants with CEMS, but retrieval and chemical-conversion uncertainties produced substantial low bias [9]. Beaudry et al. fit a rise–decay model to TEMPO plumes from 64 plants and also found weaker performance for lower-emitting sources [2].

Machine learning offers an alternative by learning emission-related plume patterns from data. Huang and Wang predicted whether TROPOMI could detect power plant $NO_2$ plumes from plant, meteorological, surface, and sensor variables rather than estimating emissions from the imagery [6]. Alnaim et al. used recurrent models with TROPOMI, meteorological, and EPA data to

estimate power plant emissions, but did not test transfer to previously unseen plants [1]. Regression has also been demonstrated for other gases: U-Plume combines learned methane plume masks with imagery and wind to infer source rates, while Dumont et al. trained a CNN on simulated $CO_2$ and wind fields to estimate power plant emissions [3, 4]. To our knowledge, no published study has used sequences of hourly TEMPO $NO_2$ imagery to infer facility-level emissions or behavior.

Estimating absolute hourly emissions from individual TEMPO scenes requires separating weak facility signals from background pollution and resolving uncertainties in retrieval, atmospheric chemistry, and transport. We instead use TEMPO's hourly sampling to detect relative changes over short observation sequences, reducing dependence on absolute column values and persistent background levels.

Our contributions are as follows:

- We formulate satellite-based power plant monitoring as a three-class change-detection task that identifies increasing, steady, and decreasing emissions behavior.
- We introduce a physics-guided deep learning framework that combines sequences of TEMPO $NO_2$ observations with meteorology and transport information. The model jointly predicts emissions behavior and a spatial map of emissions contributions within the observed scene.
- We evaluate the model on held-out power plants across geographic regions, seasons, and operating profiles and compare its performance with the temporal MLP.

## 2 Data Sources

### 2.1 TEMPO $NO_2$ Data

NASA distributes TEMPO $NO_2$ retrievals in two formats: Level 2 data on the instrument's native measurement footprints and Level 3 data regridded from each east-to-west scan [10]. We downloaded version 4 Level 2 $NO_2$ retrievals from August 2023 through August 2026. Along with $NO_2$ TVCDs, the product provides metadata including native footprint geometry, mirror-step observation times, main data quality flags, and effective cloud fractions, which we later use for regridding, quality filtering, and temporal alignment with CEMS records.

### 2.2 CEMS Records

We downloaded hourly emissions records for individual power plant units and annual facility and unit attributes from the U.S. Environmental Protection Agency's Clean Air Markets Program Data (CAMPD) APIs for August 2023 through September 2026. From the hourly records, we use facility and unit identifiers, reporting date and hour, $NO_x$ mass and its measurement flag, operating time, and primary fuel. From the annual attributes, we use facility coordinates and unit-level primary fuel.

### 2.3 HRRR Weather Data

The High-Resolution Rapid Refresh (HRRR) is an hourly weather model with a 3 km grid covering the contiguous United States. From NOAA's public HRRR archive, we downloaded hourly 2 m air temperature and eastward and northward wind components at 80 m above ground for August 2023 through August 2026.

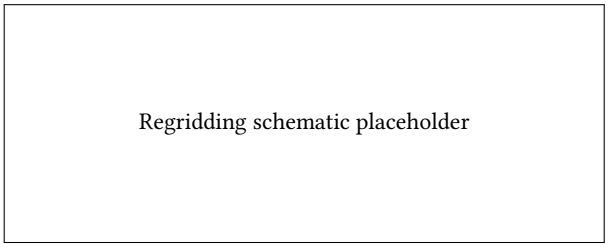



**Figure 1: TEMPO Level 2 regridding to the AOI grid.**

## 3 Methods

### 3.1 AOI Construction

For each power plant, we define a 72 km × 72 km area of interest (AOI) centered on the facility and divide it into a 24 × 24 grid of 3 km × 3 km cells. For each AOI-hour, we sum the unit-level $NO_x$ mass reported by all facilities in the AOI and discard the hour if any contributing measurement is invalid or unavailable. We score each AOI using its median total $NO_x$ mass across hours whose mean unit operating time meets or exceeds the AOI-specific median. We retain the highest-scoring 50% to focus the dataset on emissions signals more likely to be detected by TEMPO.

### 3.2 TEMPO Regridding and Quality Filtering

TEMPO Level 2 observations lie on irregular native footprints, whereas our model uses a fixed spatial grid. We project the footprint boundaries and AOI grid to the contiguous-U.S. Albers equal-area coordinate system. We then assign each AOI grid cell an $NO_2$ TVCD by averaging the TEMPO footprints that intersect it, weighted by their area of overlap. Figure 1 illustrates this mapping from native footprints to the fixed AOI grid.

A footprint contributes to the $NO_2$ average only when its main data quality flag equals zero and its effective cloud fraction does not exceed 0.20. Grid cells without an accepted footprint overlap remain masked, and we retain a binary mask of the original valid cells. We fill the masked $NO_2$ cells by bilinear interpolation before later processing.

### 3.3 Sample Construction

Each sample contains five consecutive observations of one AOI, indexed from $t_0$ to $t_4$. We retain a sequence only when each pair of adjacent TEMPO scans is separated by 40–70 minutes and every $NO_2$ raster has valid pixel coverage of at least 85%. At each timestep, we pair the TEMPO raster with the HRRR temperature and wind rasters from the closest hour interpolated to a common 24 × 24 raster size and spatial coverage as the $NO_2$ rasters.

Because CEMS records correspond to clock hours rather than TEMPO scan times, we linearly interpolate the AOI-hour $NO_x$ emissions between the surrounding hourly CEMS records at each scan time. Let $E_i$ denote the interpolated emissions at time $t_i$. For

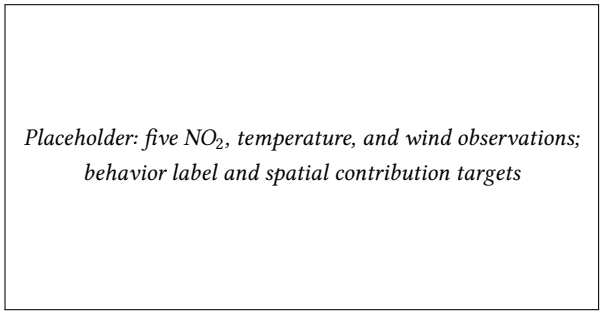



**Figure 2: Example five-timestep sample.**

scans $t_a$ through $t_b$, we calculate the time-weighted mean

$$\overline{E}_{a:b} = \frac{\sum_{i=a}^{b-1}(t_{i+1} - t_i)(E_i + E_{i+1})/2}{t_b - t_a}. \tag{1}$$

We define the sequence change as $\Delta E = \overline{E}_{2:4} - \overline{E}_{0:2}$ and use the threshold $\theta = 250\ \mathrm{lb\,h^{-1}}$ to assign its category:

$$c = \begin{cases} \text{increase}, & \Delta E \geq \theta, \\ \text{steady}, & -\theta < \Delta E < \theta, \\ \text{decrease}, & \Delta E \leq -\theta. \end{cases} \tag{2}$$

Each sample also includes two $24\times24$ spatial contribution targets so the model can jointly learn plume attribution. We interpolate each facility's emissions to the five scan times and calculate its change using the same early and late windows as the class label. For increase samples, we normalize positive facility changes and distribute them around the facility locations; decrease samples use the normalized magnitudes of negative changes. The unused target is an all-zero raster, and steady samples use all-zero rasters for both targets. Figure 2 summarizes one example sequence.

## 3.4 Transport Model

We also supply the model with rasters that use a transport model to separate new $NO_2$ emissions from existing plume at a given timestep. For each interval $\Delta t_i = t_{i+1} - t_i$, let $N_i(\boldsymbol{x})$ denote the earlier $NO_2$ raster, $\boldsymbol{u}_i(\boldsymbol{x})$ its wind field, and $B_i$ the median $NO_2$ value along the raster boundary. We approximate transport as

$$\begin{aligned} \overline{\boldsymbol{u}}_i &= \gamma\left[(1-\lambda)\boldsymbol{u}_i + \lambda \boldsymbol{u}_{i+1}\right], \\ A_i(\boldsymbol{x}) &= N_i(\boldsymbol{x} - \overline{\boldsymbol{u}}_i(\boldsymbol{x})\Delta t_i), \\ \sigma_i &= \sigma_0\sqrt{\frac{\Delta t_i}{1\,\mathrm{h}}}, \\ D_i &= B_i + G_{\sigma_i} * (A_i - B_i), \\ \widehat{N}_{i+1} &= B_i + \exp\left(-\frac{\Delta t_i}{\tau}\right)(D_i - B_i), \\ R_i &= N_{i+1} - \widehat{N}_{i+1}. \end{aligned} \tag{3}$$

The semi-Lagrangian step backtraces each destination cell and uses bilinear sampling to obtain the advected field $A_i$. Convolution with the Gaussian kernel $G_{\sigma_i}$ disperses the enhancement above the boundary background, and the exponential term applies effective $NO_2$ decay with lifetime $\tau$. The parameters $\gamma$, $\lambda$, $\sigma_0$, and $\tau$ control wind-speed scaling, endpoint-wind blending, dispersion, and decay; we select them using validation data as part of the model optimization procedure described later.

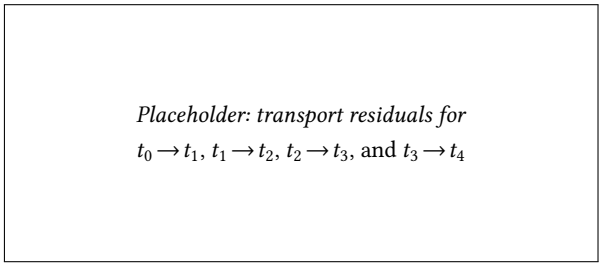



**Figure 3: Transport residuals for a five-timestep sample.**

We compute one residual $R_i$ for each adjacent pair, producing four residual rasters per sample. We transport the earlier validity mask through the same advection and dispersion steps and combine it with the later mask to identify unreliable residual pixels. Figure 3 shows the four residuals for an example sequence.

## 3.5 Dataset Stratification

Our split design has three goals: prevent leakage from overlapping AOIs, approximate a 70%/15%/15% division, and distribute the three classes across the splits while limiting class imbalance. We group overlapping AOIs into connected clusters and treat each cluster as an indivisible assignment unit. Because each cluster contains an arbitrary number of AOIs, samples, and class instances, cluster-level assignment cannot generally meet every target exactly. Steady samples account for $\sim$ 90% of the raw sample pool, compared with $\sim$ 7% increase and $\sim$ 3% decrease samples. Although we later use a class-weighted training loss, we control the composition of each split so that steady behavior does not dominate training or evaluation.

We use mixed-integer programming to assign each cluster to exactly one split while minimizing normalized deviations from the target AOI and sample counts. The first optimization targets the 70%/15%/15% division and fixes the resulting training clusters. A second optimization divides the remaining clusters equally between validation and test while balancing their AOI, sample, and per-class counts. We then downsample each training class to the size of the smallest class. In validation and test, we retain every increase and decrease sample and uniformly sample steady examples until their count matches that of the more common change class.

## 3.6 ConvGRU Model

We use a convolutional recurrent neural network with a convolutional gated recurrent unit (ConvGRU) to preserve spatial structure while combining the five observations in each sample. At each timestep, an observation encoder processes $NO_2$, the $NO_2$ quality mask, temperature, and the two wind components. A separate encoder processes each of the four transport residuals together with its quality mask. Each encoder shares weights across time and reduces the $24 \times 24$ inputs to spatial feature maps for temporal aggregation. Figure 4 summarizes the architecture.

The encoded first observation initializes the recurrent hidden state. At each later timestep, the model concatenates the current

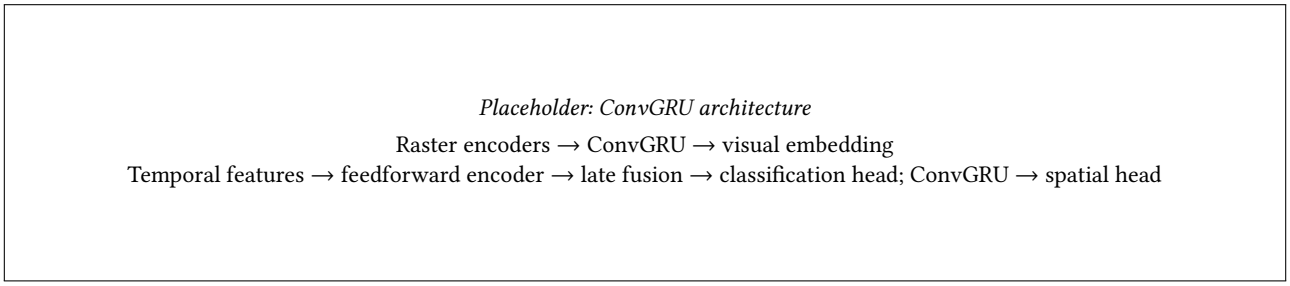



**Figure 4: ConvGRU architecture.**

observation features with the transport residual from the preceding interval, fuses them with a $1 \times 1$ convolution, and updates the hidden state with the ConvGRU. Its convolutional gates retain the spatial arrangement of plume features while learning how much information to preserve or replace across the sequence. This design lets the network use both the observed atmospheric state and the portion of each change that the transport model does not explain.

The final recurrent feature map feeds two prediction heads. For classification, we combine global average and maximum pooling to form a visual embedding. A small feedforward network separately encodes the four temporal features, and its output is concatenated with the visual embedding before the classification head produces logits for the increase, steady, and decrease classes. This late-fusion design keeps the tabular context separate while the ConvGRU learns the raster sequence. The spatial head upsamples the recurrent feature map into two $24 \times 24$ contribution rasters representing increases and decreases. Sharing the recurrent representation allows the model to learn emissions behavior and spatial attribution jointly.

## 3.7 Temporal MLP

The temporal MLP measures how much emissions-change behavior can be predicted from recurring daily and annual patterns without satellite or weather inputs. We represent each sample using four cyclic features: the sine and cosine of local solar hour (midpoint of timesteps) and the sine and cosine of day of year. The temporal MLP maps these features through hidden layers of 16 and 8 units, each followed by layer normalization and a SiLU activation. A linear output layer produces logits for the three behavior classes. Figure 5 shows this architecture.

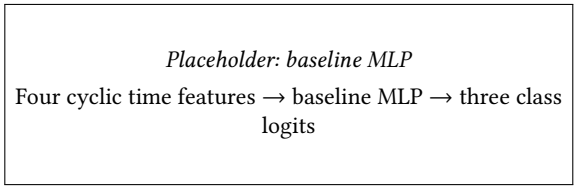



**Figure 5: Temporal MLP architecture.**

**Table 1: Dataset composition by split and class.**

| Split | AOIs | Decrease | Steady | Increase | Total |
|---|---|---|---|---|---|
| Training | 469 | 17,130 | 23,289 | 29,888 | 70,307 |
| Validation | 101 | 3,198 | 8,589 | 10,411 | 22,198 |
| Test | 97 | 3,184 | 7,764 | 10,093 | 21,041 |

**Table 2: Training configuration.**

| Parameter | Baseline MLP | ConvGRU |
|---|---|---|
| Trainable parameters | 291 | 865,805 |
| Optimizer | AdamW | |
| Batch size | 128 | 128 |
| Initial learning rate | $10^{-3}$ | $10^{-4}$ |
| Weight decay | $10^{-3}$ | $10^{-3}$ |
| Dropout | – | 0.30 |
| Spatial-loss weight $\lambda_{sp}$ | – | 1.0 |
| Gradient-norm limit | 5.0 | |
| LR scheduler | Reduce on plateau | |
| LR reduction factor | 0.5 | |
| Scheduler patience | 10 epochs | |
| Selection metric | Validation macro-F1 | |
| Early-stopping patience | 10 epochs | |
| Hardware | 1 NVIDIA RTX A5000 GPU | |

## 3.8 Training Procedure

Table 1 summarizes the final dataset used to train and evaluate both models. We standardize the $NO_2$, wind, temperature, and transport-residual rasters and the tabular inputs using the corresponding training-set mean and standard deviation. We apply these training-derived statistics unchanged to the validation and test splits.

Both classifiers use inverse-frequency class weights computed from the final training split. The temporal MLP minimizes weighted cross-entropy. For the ConvGRU, we combine this classification objective with a debiased unbalanced Sinkhorn divergence between the predicted and target contribution rasters:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{sp}} \mathcal{L}_{\mathrm{sp}}, \tag{4}$$

where $\lambda_{\mathrm{sp}}$ controls the contribution of the spatial task. The unbalanced spatial loss penalizes misplaced contribution mass while allowing the all-zero targets used by steady samples and inactive channels.

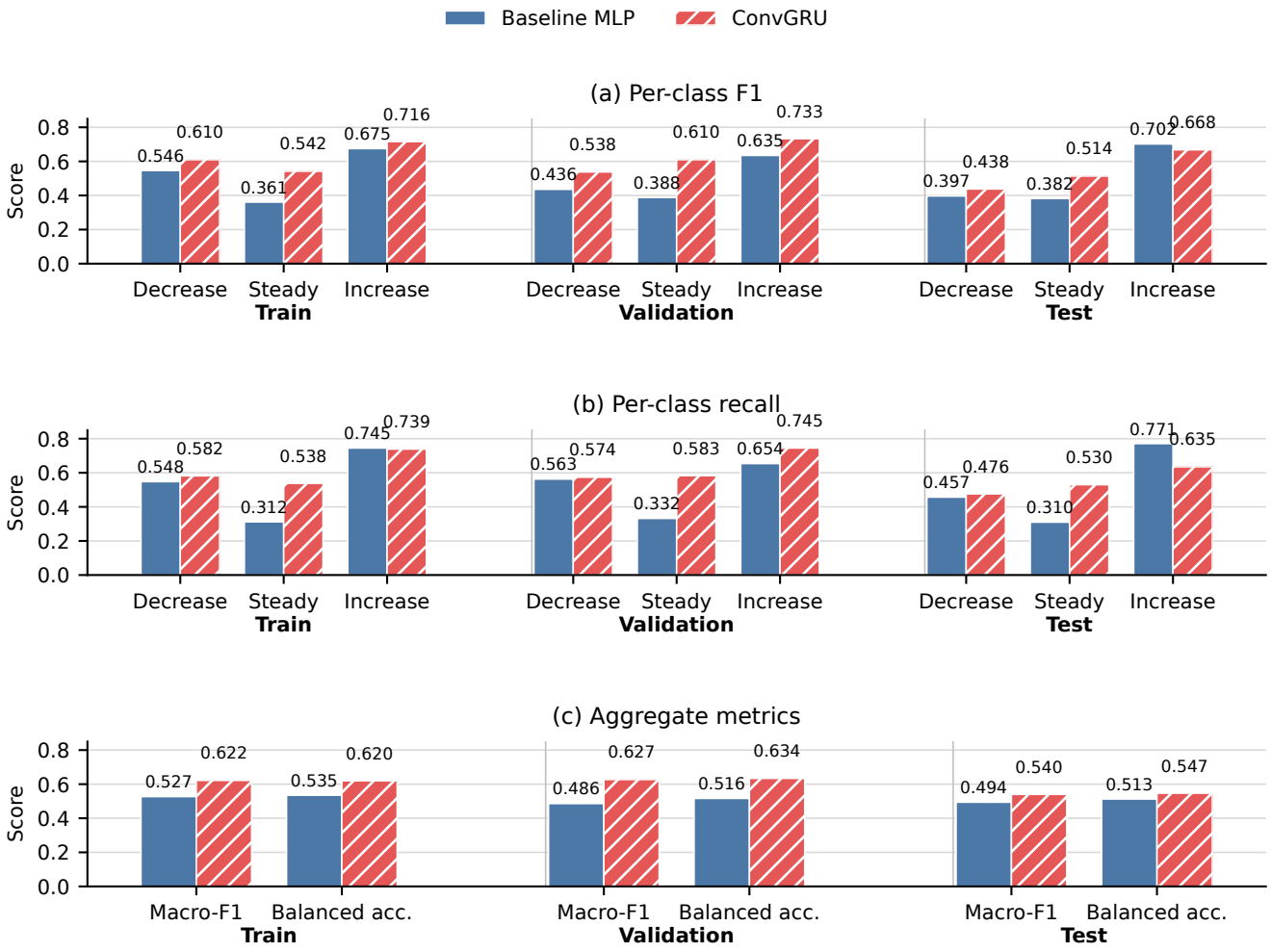



**Figure 6: Classification performance by split.**

# 4 Results

## 4.1 Classification Performance

We evaluate classification using macro-F1 and balanced accuracy to account for differences in class prevalence. On the test set, the ConvGRU achieves a macro-F1 of 0.540 and balanced accuracy of 0.547, compared with 0.494 and 0.513 for the temporal MLP. As shown in Figure 6, the improvement is concentrated in the steady class, while performance on increase samples is lower for some metrics. Results across the training, validation, and test splits show the same aggregate advantage for the ConvGRU.

## 4.2 Spatial Attribution Performance

*Placeholder: spatial attribution loss*

Per-class loss for the training, validation, and test splits.

**Figure 7: Spatial attribution loss by class and split.**

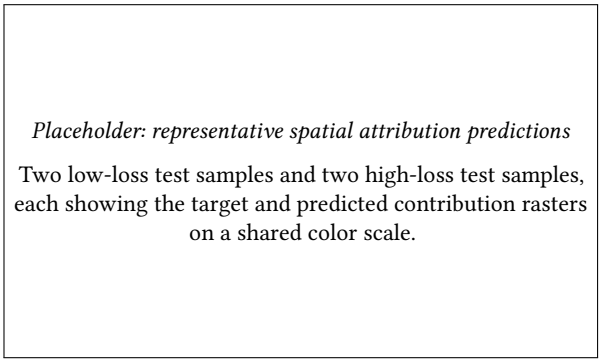



**Figure 8: Representative spatial attribution predictions.**

## 4.3 Performance Across Evaluation Strata

## 4.4 Ablation Study

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